Initial value problems

In the last lesson about linear differential equations, all the general solutions we found contained a constant of integration, C. But we're often interested in finding a value for C in order to generate a particular solution for the differential equation.

This applies to linear differential equations, but also to any other form of differential equation. The information we'll need in order to find C is an **initial condition**, which is the value of the solution at a specific point.

Only one solution will satisfy the initial condition(s), which is why the initial condition(s) allow us to narrow down the equation of the general solution to one specific particular solution.

An initial condition for a first order differential equation will take the form

$$y(x_0) = y_0$$

and the number of initial conditions required for a given differential equation will depend on the order of the differential equation, which we'll talk more about later.

An **initial value problem (IVP)** is a differential equations problem in which we're asked to use some given initial condition, or set of conditions, in order to find the particular solution to the differential equation.

Solving initial value problems

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In order to solve an initial value problem for a first order differential equation, we'll

1. Find the general solution that contains the constant of integration C.

2. Substitute the initial condition, $x = x_0$ and $y = y_0$, into the general solution to find the associated value of C.

3. Restate the general solution, and include the value of C found in step 2. This will be the particular solution of the differential equation.

Let's work through an example so that we can see these steps in action.

Example

Solve the initial value problem if $y(0) = -5$ in order to find the particular solution to the differential equation.

$$\frac{dy}{dx} + 2y = 4e^{-2x}$$

In the previous lesson, we used the integrating factor to find the general solution to this differential equation, and it was

$$y = \frac{4x + C}{e^{2x}}$$

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Once we have this general solution, we recognize from the initial condition $y(0) = -5$ that $x = 0$ and $y = -5$, so we'll plug these values into the general solution,

$$-5 = \frac{4(0) + C}{e^{2(0)}}$$

and then simplify this to solve for C.

$$-5 = \frac{0 + C}{1}$$

$$C = -5$$

So the particular solution to the differential equation is

$$y = \frac{4x - 5}{e^{2x}}$$

We should realize from this last example that the general solution always remains the same, but the particular solution changes based on the initial condition we use. The particular solution we generated here was based on $y(0) = -5$, but we would have found a different particular solution for $y(0) = 2$, and yet another for $y(0) = 3$.

And as we said before, initial value problems can be used for all types of differential equations.

This last example was an initial value problem for a linear differential equation, but be on the lookout for initial value problems in the future as we look at lots of other types of differential equations.

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